MACHINE LEARNING

Classification Metrics

Confusion Matrix · Precision · Recall · F1 Score · Macro & Weighted Average

A visual guide to evaluating your ML models • linkedin.com/learning

Confusion Matrix

What is it?

A table that compares **actual vs predicted** values from a classification model.

It reveals not just accuracy — but

HOW and WHERE your model makes errors.

	Predicted YES	Predicted NO
Actual YES	**TP** True Positive	**FN** False Negative
Actual NO	**FP** False Positive	**TN** True Negative

✅ Green = correct · 🔴 Red = errors

Why Not Just Use Accuracy?

Accuracy alone can be deceiving. Here's why the Confusion Matrix matters:

🎯

Full Performance Picture

See exactly which classes are being confused, not just a single number.

⚠️

Understand Error Types

Distinguish between False Positives & False Negatives — they have very different costs.

⚖️

Handle Imbalanced Data

A model that predicts 'No' 99% of the time looks 99% accurate — but it's useless.

💡 *Confusion Matrix = the foundation of all other classification metrics*

🎯 Precision

$$\text{Precision} = TP \div (TP + FP)$$

What it measures

Of all the positives your model predicted, how many were actually correct?

Focuses on quality of positive predictions.

Real-world example

📧 Spam filter:

If 10 emails flagged as spam, and 8 were truly spam:

Precision = 8 ÷ 10 = 0.80 (80%)

⚡ *Use precision when FALSE POSITIVES are costly (e.g. medical diagnosis, spam filters)*

📡 Recall (Sensitivity)

$$\text{Recall} = TP \div (TP + FN)$$

What it measures

Of all actual positives in the dataset, how many did your model successfully catch?

Focuses on coverage / completeness.

Real-world example

🏥 Cancer detection:

If 10 actual cancer cases, model identified 8:

Recall = 8 ÷ 10 = 0.80 (80%)

⚡ *Use recall when FALSE NEGATIVES are costly (e.g. disease detection, fraud prevention)*

Precision vs Recall — The Trade-off

You often can't maximise both at the same time.

🎯 HIGH PRECISION

Low Recall

- Model is very conservative
- Only predicts positive when very sure
- Misses many true positives (high FN)
- Good for: spam filters, ad recommendations

📡 HIGH RECALL

Low Precision

- Model is aggressive in flagging positives
- Catches most true positives (low FN)
- Many false alarms (high FP)
- Good for: disease detection, fraud alerts

💡 *This is exactly why F1 Score exists — it balances both!*

⚖️ F1 Score — The Balancer

$$F1 = 2 \times (Precision \times Recall) \div (Precision + Recall)$$

🔗

Harmonic Mean

F1 uses harmonic mean, not arithmetic. This punishes extreme imbalance between Precision and Recall.

⚠️

When it helps

Best when you need a single metric and both false positives & false negatives matter equally.

📊

Range

F1 ranges from 0 to 1.

1.0 = perfect score

0.0 = total failure

>0.7 = generally good

F1 is your go-to metric for imbalanced datasets | It won't let a great Precision hide a poor Recall

Macro vs Weighted Average

For multi-class problems, you need an averaging strategy

🏛️ Macro Average

WHAT: Calculate the metric for each class separately, then take a simple unweighted average.

WHEN: Use when all classes are equally important regardless of size.

```
Class A: 0.9 · Class B: 0.7 · Class C: 0.5
Macro Avg = (0.9 + 0.7 + 0.5) ÷ 3 = 0.70
```

⚠️ *Can be misleading if class sizes differ greatly*

⚖️ Weighted Average

WHAT: Calculate metric per class, then average weighted by the number of samples (support) in each class.

WHEN: Use for imbalanced datasets — larger classes influence the average more.

```
Class A (n=100): 0.9 · Class B (n=30): 0.7
Weighted = (100×0.9 + 30×0.7) ÷ 130 = 0.85
```

✅ *Better reflects real-world performance on skewed data*

scikit-learn gives you both in classification_report() | Always check both!

📋 Quick Reference Cheat Sheet

Metric	Formula	Focus	Best Used When
Confusion Matrix	TP / TN / FP / FN table	Full error picture	Always — start here
Precision	TP ÷ (TP + FP)	Quality of positive preds	When FP is costly
Recall	TP ÷ (TP + FN)	Coverage of actual positives	When FN is costly
F1 Score	2 × P × R ÷ (P + R)	Balance of P & R	Imbalanced datasets
Macro Avg	Mean of per-class metrics	Equal class weight	Balanced classes
Weighted Avg	Support-weighted mean	Sample-proportional	Imbalanced classes

Key Takeaways 🚀

01

Always start with the Confusion Matrix

It's the source of truth for all other metrics.

02

Precision vs Recall is a trade-off

Choose based on the cost of FP vs FN in your use case.

03

F1 Score balances both

Use it when you can't sacrifice either precision or recall.

04

Pick the right average

Macro for equal class importance · Weighted for imbalanced data.

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